OFFICIAL INSTITUTIONAL BLUEPRINT

Scholar SMS & LMS Suite

Enterprise-Grade Modular School Management System & CBT Digital Campus

> A fully comprehensive operational, pedagogical, and technical executive overview detailing multi-role workflows, digital examination terminals, dual-grading models, attendance safeguards, and omnichannel communications.

DOCUMENT REFERENCE
SCH-DOC-EXEC-2026-V2

TARGET AUDIENCE
Academic Boards, Executive Directors & IT Steering

OPERATIONAL ROLES COVERED
Super Admin, Faculty, Student, Guardian, Public

PRIMARY TECHNOLOGICAL ARCHITECTURE
Modular Monolith, RBAC & SHA-256 Chained Ledger

EXAMINATION SYSTEM (CBT)
5 Question Types, Countdown & Live Auto-Save

PUBLICATION DATE
Academic Year 2026/2027

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1. Strategic Vision & Executive Summary

STRATEGIC CORE

Scholar SMS represents an enterprise-grade, integrated institutional solution engineered to eliminate administrative fragmentation in modern primary, secondary, and tertiary educational organizations. Traditionally, institutions grapple with disparate software packages for student registration, staff payroll, daily roll call, online examinations, report card generation, and guardian communication. Scholar SMS consolidates these disparate functions into a unified, secure, high-performance ecosystem.

5	100%	4	SHA-256
AUTHORIZED USER ROLES	AUTOMATED GRADING SYNC	COMMUNICATION GATEWAYS	CRYPTOGRAPHIC AUDIT

Core Institutional Objectives

- **Pedagogical Continuity:** Seamless bridging of physical classroom instruction with digital learning materials, interactive computer-based assessments, and instant solution reviews.
- **Uncompromising Governance:** Cryptographically verifiable audit trails where every grade modification, attendance seal, and enrollment action is immutably linked in a chained ledger.
- **Parental Transparency:** Real-time automated visibility for guardians into their ward's academic trajectory, attendance punctuality, behavioral notes, and fee standing.
- **Administrative Efficiency:** Automated holiday calendars ensuring that national and school-declared recesses never skew daily attendance analytics or report card metrics.

2. Role-Segregated Operational Ecosystem

GOVERNANCE & RBAC

The platform operates on strict Role-Based Access Control (RBAC). User interfaces, navigation hierarchies, and mutation endpoints are strictly segregated to prevent privilege escalation or data leakage across personas.

Operational Role	Core Workspace & Responsibilities	Key System Capabilities
SUPER ADMINISTRATOR	Executive oversight of all institutional domains, human resources, system configuration, academic calendars, and audit ledger integrity.	Student bulk Excel onboarding, staff onboarding, holiday calendars, dual-grading policy setup, API gateway credentials, audit verification.
FACULTY / INSTRUCTOR	Curriculum execution, student continuous assessments, roll-call attendance, LMS course authoring, and examination grading.	Course syllabus authoring, digital material distribution, Q&A question studio authoring, rubric essay scoring, QR roll-call scanning.
STUDENT CANDIDATE	Digital classroom engagement, learning material consumption, interactive CBT testing, and attendance monitoring.	Enrolled course access, interactive timed examination terminal, real-time question palette, immediate score reports, personal 360 profile.
GUARDIAN / PARENT	Family oversight of registered wards, progress tracking, attendance safety notifications, and administrative advisories.	Multi-ward toggle, real-time arrival/absence alerts, terminal report card inspection, faculty comments, campus announcements.
PUBLIC VISITOR	Public community discovery, curriculum inquiries, event exploration, and multimedia campus showcases.	Dynamic course catalog, campus life media gallery, institutional event calendar, digital admission inquiry terminal.

3. Modern LMS Tests, Quizzes & Examinations Suite

PEDAGOGICAL TECHNOLOGY

The LMS suite incorporates a comprehensive Computer-Based Testing (CBT) engine designed to support high-stakes formal examinations (Midterms, Final Semesters) as well as continuous formative assessments (Weekly Quizzes, Practice Drills).

STUDIO Q&A Authoring Studio

An intuitive workspace enabling faculty to author questions across **5 distinct formats**:

- **Single-Choice MCQ:** Standard choice with exactly one validated correct option.
- **Multi-Select MCQ:** Multi-answer selection requiring all correct choices for credit.
- **True / False:** Dedicated tactile truth-value validation.
- **Short Answer:** Pattern and comma-separated keyword matching engine.
- **Long Essay:** Subjective analytical submissions reserved for instructor rubric review.

CBT Student Examination Terminal

A distraction-free exam player built for strict focus and reliability:

- **Synchronized Countdown:** Real-time timer with server-backed duration checks.
- **Question Palette:** Visual matrix showing answered, current, and unanswered items.
- **Asynchronous Auto-Save:** Background auto-saving prevents data loss on network drops.
- **Anti-Cheating Safeguards:** Question order randomization and choice shuffling.
- **Zero-Timer Enforcement:** Automatic answer freeze and submission when time expires.

GRADING Rubric Review & Submissions Console

Centralized evaluation interface displaying candidate attempt rosters, submission timestamps, and automated scores. Provides dedicated mark-allocation inputs and qualitative feedback fields for subjective essay prompts.

REPORT Candidate Performance Report Card

Post-submission diagnostic report providing total marks, percentage, pass/fail status, duration taken, attempt counting, and solution explanations (when permitted by instructor policy).

REPOSITORY Question Banks & Reusable Item Libraries

Centralized question banking system categorizing items by subject, grade level, and difficulty. Teachers and administrators can author standalone banks, export complete assessments directly into bank archives, and selectively clone questions from banks into multiple tests, midterms, and finals without re-typing.

INGESTION Bulk Excel / CSV Question Onboarding Engine

Automated bulk ingestion workflow with standard downloadable spreadsheet templates. Faculty can prepare dozens of multi-choice, true/false, short answer, or essay questions offline and import them in one click with automated validation, option mapping, and total point recalculation.

Automated Timed Access Gates & Publishing Enforcement

To preserve academic integrity, assessments enforce strict chronological and lifecycle controls. Unreleased assessments in *Draft Mode* remain invisible to students. Published assessments are governed by synchronized **Start Time** and **End Time** gates: premature attempts are held behind locked doors displaying the exact opening time, while expired assessments strictly prohibit candidate entry. Active sessions are automatically bounded so that no candidate can exceed the designated institutional closing deadline.

4. Dual-Grading Scheme & Holiday Calendar Safeguards

ACADEMIC ARCHITECTURE

To bridge digital learning with traditional physical classroom activities, Scholar SMS pioneers an integrated **Dual-Grading Matrix** paired with an intelligent **Academic Calendar Engine**.

Dual-Grading Framework (Continuous LMS + Classroom Offline)

Recognizing that education occurs both in digital labs and physical lecture halls, the platform allows academic administrators to configure custom evaluation weighting per course:

Grading Component	Standard Allocation	Data Source & Evaluation Channel
Online Digital LMS	**30% – 50%**	Interactive CBT Quizzes, Module Knowledge Checks, Automated Objective Grading.
Offline Classroom Coursework	**50% – 70%**	Physical Midterms, Laboratory Practicals, Written Final Papers, Oral Presentations.
Terminal Synthesis	**100%**	Automated weighted composite calculated dynamically on Official Terminal Report Cards.

Intelligent Calendar & Attendance Safeguards

The academic calendar module registers **National Holidays**, **Regional Holidays**, and **Institutional Non-Instructional Days**. The daily attendance roll call engine cross-references this calendar automatically—ensuring that students are never recorded as absent during school breaks and overall institutional attendance percentages remain accurate.

5. Omnichannel Communications & Gateway Engine

INSTITUTIONAL CONNECTIVITY

Scholar SMS features an enterprise multi-channel notification engine capable of instant and scheduled broadcasting across four critical communications vectors:

SMTP & Transactional Email

Full configuration of institutional mail transfer agents with support for custom host, port, authentication credentials, and modern encryption standards (TLS/SSL).

SMS Gateway Integration

Direct carrier integration supporting global telecom providers (Twilio, Infobip) for mission-critical notifications such as unexcused absence alerts and emergency announcements.

WhatsApp Cloud API

Native integration with Meta's WhatsApp Cloud API infrastructure, enabling conversational parent updates, academic reminders, and automated bulletin delivery directly to parent handsets.

Broadcast Template Engine

Curated template repository for religious, seasonal, gender-specific, and emergency greetings with automated date-triggered sending capabilities.

6. Academic Registry, Security & Cryptographic Audit

INTEGRITY & GOVERNANCE

SIS Student Information System

- Automated institutional admission numbering (e.g. `SCH-2026-001`).
- Cohort allocation and guardian linkage.
- Bulk Excel onboarding with automated column validation.
- Student 360 profile tracking academic, attendance, and disciplinary histories.

HR Faculty & Staff HR Management

- Automated employee code generation (e.g. `EMP-2026-001`).
- Departmental assignment and academic qualification logging.
- Form-teacher classroom leadership and subject accreditation.
- Compensation tracking and active employment status toggles.

Cryptographically Chained (SHA-256) Audit Trail

Scholar SMS guarantees institutional transparency through a tamper-evident cryptographic ledger modeled on Merkle chaining principles. Every significant administrative action—including student enrollment, grade modification, attendance sealing, and broadcast dispatch—is cryptographically hashed with the preceding block's hash:

```
Current_Hash = SHA-256(Previous_Hash + Actor_ID + Action_Type + Entity_Payload + Timestamp)
```

An interactive **Audit Verification Console** allows administrators and external auditors to recalculate the chain from the Genesis block to the tip in real time, detecting unauthorized database mutations instantly.

7. Public Engagement Showcase & Dynamic CMS

MARKETING & COMMUNITY

Complementing the private portal workspaces, Scholar SMS includes a public web landing page built on the Scholar v1.0.0 aesthetic framework:

- **Dynamic Campus Events Showcase:** Administrative CRUD enabling real-time publication of symposiums, athletic meets, and open days directly to the public portal without code updates.
- **Categorized Campus Media Gallery:** Image upload and categorization engine allowing prospective parents and visitors to explore campus facilities, STEM laboratories, and ceremonies.
- **Public Course Curriculum Catalog:** Prospective students can explore available academic subjects, syllabi summaries, and instructor credentials.
- **Interactive Inquiries Desk:** Direct admission inquiry form capturing prospective guardian details directly into the administrative CRM.
- **User Profile Avatar Engine:** Self-service profile image management across all authenticated user accounts, ensuring personalized dashboards.

8. Role Capability Matrix

SYSTEM FUNCTIONAL MATRIX

Functional Module	Super Admin	Faculty / Teacher	Student	Guardian / Parent
Executive Dashboard & Analytics	✓ (Full System)	✓ (Assigned Cohort)	✓ (Personal)	✓ (Linked Wards)
Student Enrollment & Bulk Excel Upload	✓	✗	✗	✗
Faculty Onboarding & Staff Directory	✓	✓ (Directory Read)	✗	✗
Academic Terms & Holiday Calendar Setup	✓	✗	✗	✗
LMS Course Curriculum & File Materials	✓	✓	✓ (Download)	✓ (Overview)
CBT Question Authoring Studio (5 Types)	✓	✓	✗	✗
Question Banks & Reusable Item Repositories	✓	✓	✗	✗
Bulk Assessment Upload (Excel / CSV Template)	✓	✓	✗	✗
Timed Assessment Access Gates & Publishing Control	✓ (Configure)	✓ (Configure)	✓ (Within Window)	✗
Interactive Examination Player Terminal	✓ (Preview)	✓ (Preview)	✓ (Take Exam)	✗
Teacher Rubric Essay Grading Console	✓	✓	✗	✗
Dual-Grading Matrix & Terminal Report Cards	✓	✓	✓ (View Own)	✓ (View Ward)
Roll-Call & Contactless QR Scanner	✓	✓	✗	✗
Omnichannel Gateways (SMTP/SMS/WhatsApp)	✓	✗	✗	✗
Public Events & Gallery CMS	✓	✗	✗	✗
Cryptographic SHA-256 Audit Verification	✓	✗	✗	✗

9. Institutional Sign-Off & Operational Readiness

CERTIFICATION

The **Scholar SMS & LMS Suite** has completed end-to-end unit, integration, and user-acceptance validation across all functional domains. The application stands in a state of full operational readiness for immediate institutional deployment, onboarding, and semester commencement.

EXECUTIVE ACADEMIC BOARD APPROVAL

Office of the Academic Registrar

Scholar Educational Technologies

TECHNICAL VERIFICATION & SYSTEMS SIGN-OFF

Lead Systems Architect

Enterprise Systems Group